US Traffic Accident Hotspot & Risk Factor Analysis

Case Study

Traffic accidents remain one of the leading public safety challenges worldwide. Large volumes of traffic, urban congestion, environmental conditions, and road infrastructure collectively contribute to accident occurrences.

Transportation authorities and city planners need data-driven insights to understand where accidents occur most frequently and what factors influence accident severity. However, traffic accident datasets are extremely large and complex, making manual analysis inefficient.

This project analyzes large-scale US accident data to identify patterns, understand risk factors, and highlight accident hotspots across the country.

Business Objective

The goal of this project is to analyze traffic accident data to identify patterns that contribute to accident occurrences and severity.

Key objectives include:
- Identify geographic accident hotspots.
- Understand how time patterns influence accident frequency.
- Analyze the impact of weather conditions on accidents.
- Investigate how road infrastructure affects accident occurrences.
- Provide data-driven recommendations for improving road safety.

Dataset Information

Dataset: US Accidents Dataset (2016–2023)
Source: Kaggle

Dataset Characteristics:
- Coverage: 49 US states
- Time Period: February 2016 – March 2023
- Records: ~7.7 million accidents
- Features: ~46 variables
- File Format: CSV

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Dataset Credit:

- Moosavi, S., Samavatian, M. H., Parthasarathy, S., & Ramnath, R. (2019). A Countrywide Traffic Accident Dataset.
- Moosavi, S., Samavatian, M. H., Parthasarathy, S., Teodorescu, R., & Ramnath, R. (2019). Accident Risk Prediction based on Heterogeneous S

Key attributes include accident severity, timestamps, weather conditions, road infrastructure indicators, and geographic coordinates.

Tools & Technologies Used

Python – Data analysis and processing
Pandas – Data manipulation and preprocessing
NumPy – Numerical computations
Matplotlib – Data visualization
Seaborn – Statistical visualizations
Folium – Geospatial heatmaps
Jupyter Notebook – Development environment

Project Roadmap

This project follows the standard Data Analysis lifecycle:

1. Ask – Define key questions related to accident patterns.
2. Prepare – Load and inspect the dataset.
3. Process – Clean and preprocess the data.
4. Analyze – Perform exploratory data analysis (EDA).
5. Share – Create visualizations to communicate insights.
6. Act – Provide recommendations for improving road safety.

Data Cleaning & Preprocessing

Several preprocessing steps were applied to prepare the dataset for analysis.

• Handling missing values using median imputation and data filtering.
• Removing irrelevant or redundant columns.
• Converting data types for memory optimization.
• Transforming timestamp columns into useful time-based features.
• Ensuring consistency across categorical variables.

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Feature Engineering

New features were derived from the dataset to enhance analysis.

Time-Based Features:
• Hour of the accident
• Day of week
• Month and season

Infrastructure Indicators:

Binary indicators were analyzed for junctions, traffic signals, railway crossings, and stop signs to understand their influence on accident frequency.

Exploratory Data Analysis

Several visualizations were created to uncover meaningful patterns in the dataset

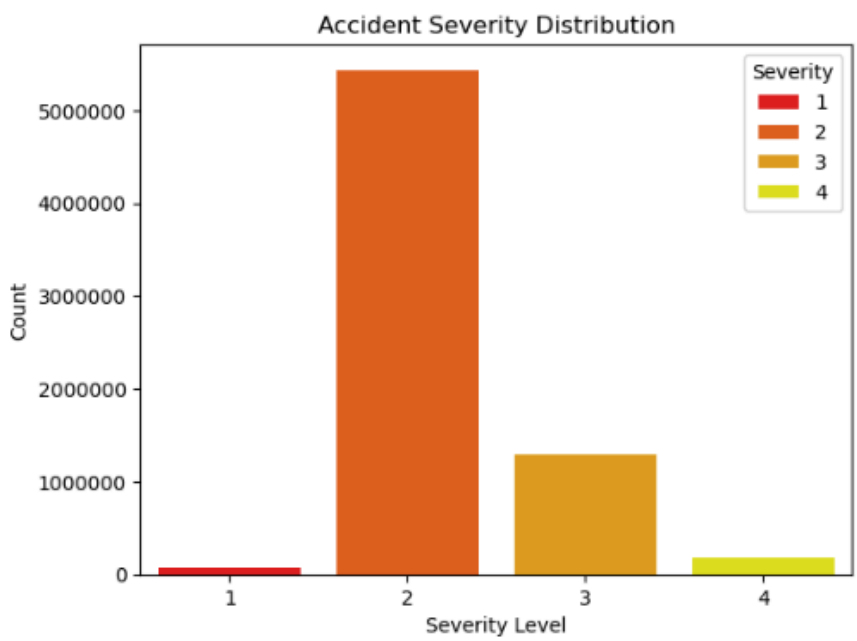

Figure 1 – Accident Severity Distribution

Key Insights:

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1. It is observed that the majority of accidents fall under Severity Level 2, which indicates that most accidents in the dataset are moderately serious but not extremely severe.

2. It is also observed that high severity accidents Level 3 and Level 4 occur less frequently, suggesting that severe collisions are rare, but still represent safety concerns.

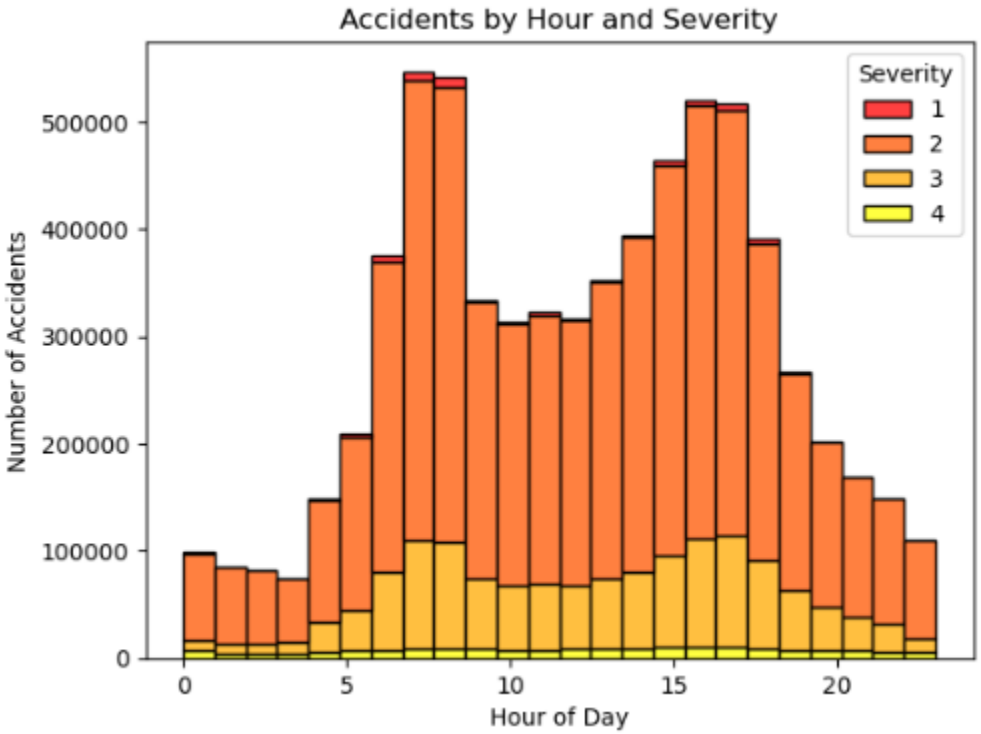

Figure 2 – Accident Frequency by Hour of Day

Key Insights:

1. Accident frequency rises sharply between 6–9 a.m., peaking around 7–8 a.m., indicating increased traffic during morning work commutes. A second peak occurs between 3–6 p.m., especially around 4–5 p.m., during evening return traffic.

2. Most accidents fall under **Severity Level 2 (moderate collisions)**, but **Severity Levels 3 and 4 increase during peak hours**, likely due to traffic congestion, driver stress, and higher vehicle interaction.

3. Accident counts are **significantly lower between late morning and early afternoon**, and during **night hours**, reflecting reduced traffic volume outside major commuting period

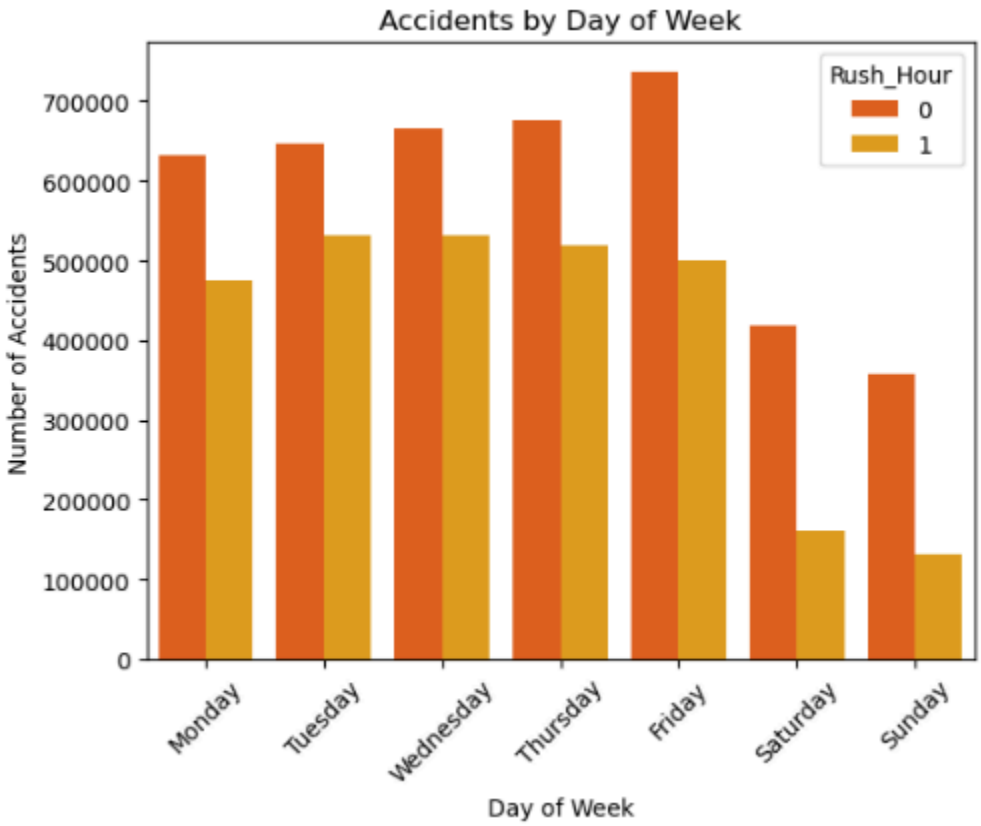

Figure 3 – Accidents by Day of Week

Key Insights:

1. Accident counts gradually increase from Monday to Friday, with Friday recording the highest number of accidents, likely due to heavy work commuting and increased weekend travel activity.

2. Accident numbers **drop significantly on weekends** because of reduced work commuting, yet **non-rush hours still account for a large share of accidents**, indicating accidents occur throughout the day, not just during peak traffic times.

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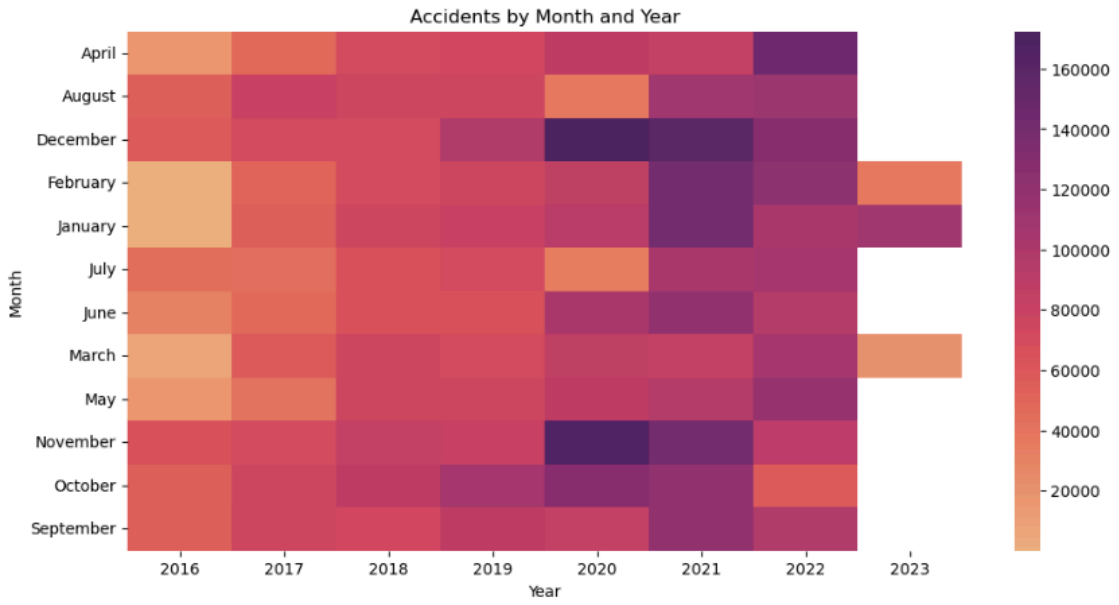

Figure 4 – Accidents by Month Vs Year

Key Insights:

1. From 2016 to 2018, accident numbers slowly increased, with noticeable spikes during summer months (May–July) when travel activity is usually higher.
2. Drop During COVID-19 Restrictions: In early 2020, accident counts declined as lockdowns and stay-at-home orders reduced road travel across many regions.
3. Increase After Reopening: As restrictions eased in 2021, accident numbers rose again, with clear spikes during November and December, likely due to increased mobility and holiday travel during Thanksgiving, Christmas, and New Year**.**

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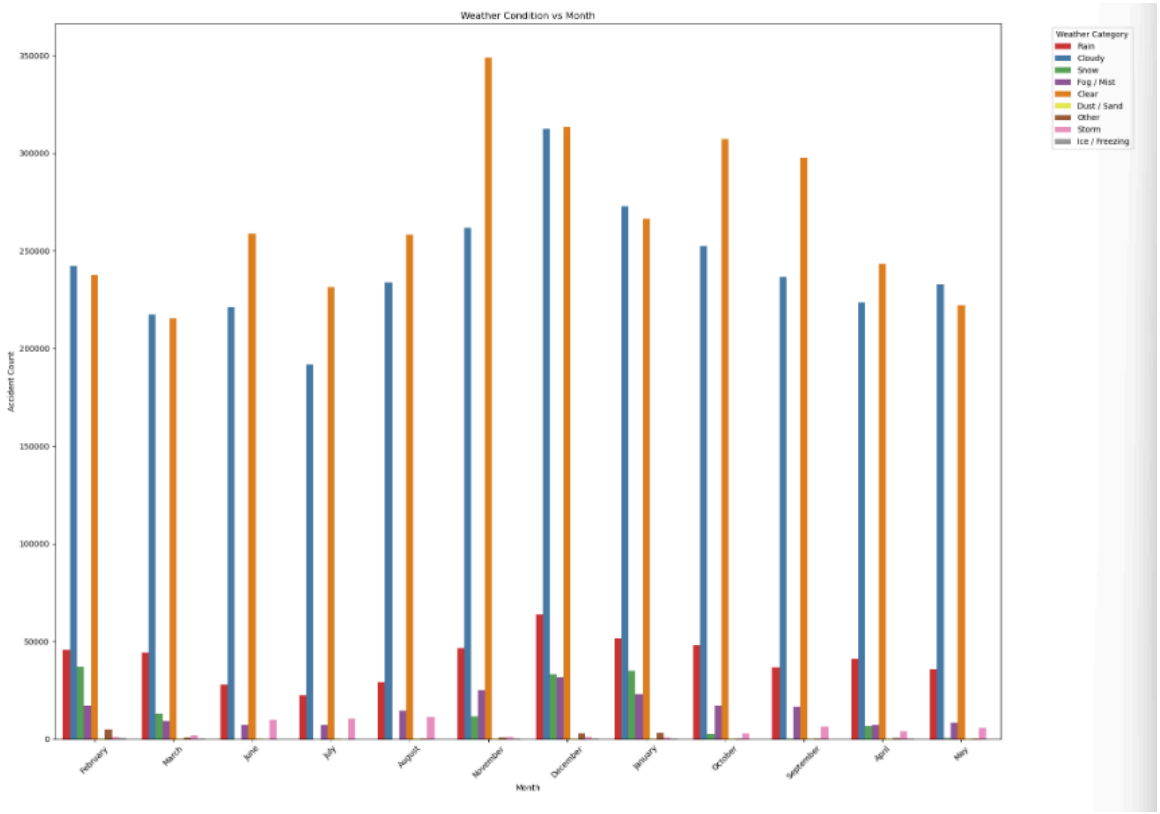

Figure 5 – Accidents by Weather Conditions

Key insights:

1. Most accidents happen during clear or cloudy weather, suggesting that accident frequency is mainly driven by higher traffic volume when conditions are good and more people are on the road**.**

2. Rain-related accidents remain consistently high, especially in late summer and early winter, as wet roads reduce tire grip and make vehicles more prone to collisions.

3. Snow-related accidents rise during winter months, particularly December and January, because slippery roads and reduced vehicle control increase accident risk, while extreme conditions like storms or dust events show fewer accidents since people tend to travel less.

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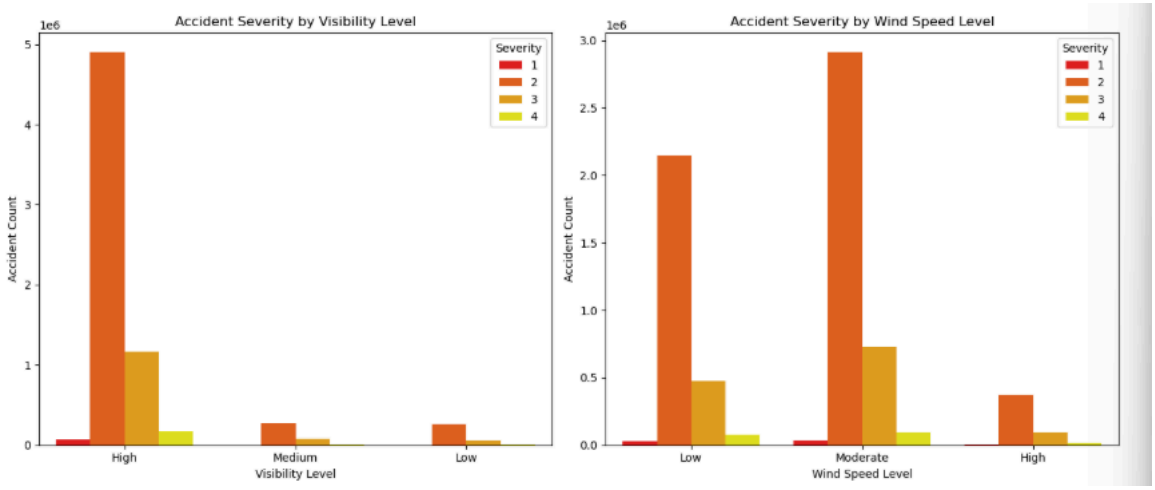

Figure 6 –Visibility vs Severity & Wind Speed vs Severity

Key insights:

1. Most accidents occur when visibility is good, suggesting that crashes are more related to higher traffic volume during normal driving conditions rather than poor visibility.

2. Across all visibility levels, Severity Level 2 accidents are the most common, indicating that most crashes are moderately severe rather than extremely serious.

3. Accidents are most frequent during moderate wind conditions, while strong winds or very low visibility show fewer accidents, likely because people drive less or more cautiously during extreme weathe

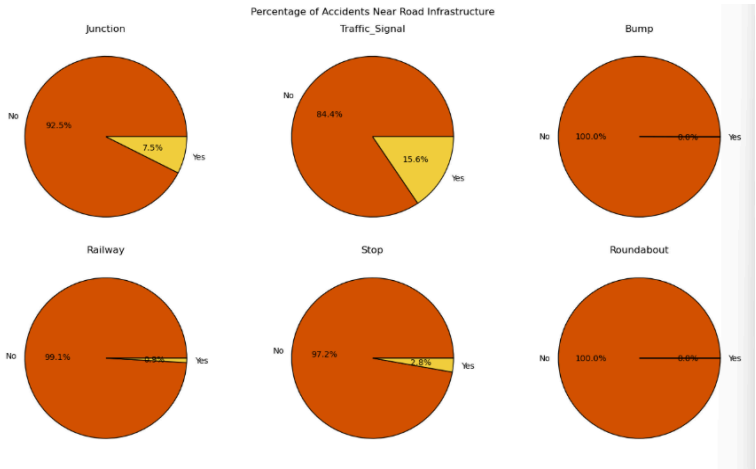

Figure 7 – Accidents Near Road Infrastructure

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Key insights:

1. Only a **small share of accidents occur near junctions**, meaning most crashes happen away from them. However, **traffic signals show the highest share among road infrastructure**, likely due to sudden stops, turning vehicles, and congestion at intersections.

2. **Speed bumps and roundabouts show almost no recorded accidents**, suggesting these features may be effective in slowing traffic and improving road safety.

3. Accidents near **railway crossings are very rare**, and only a **small fraction occur near stop signs**, indicating they are not major accident hotspots in the dataset.

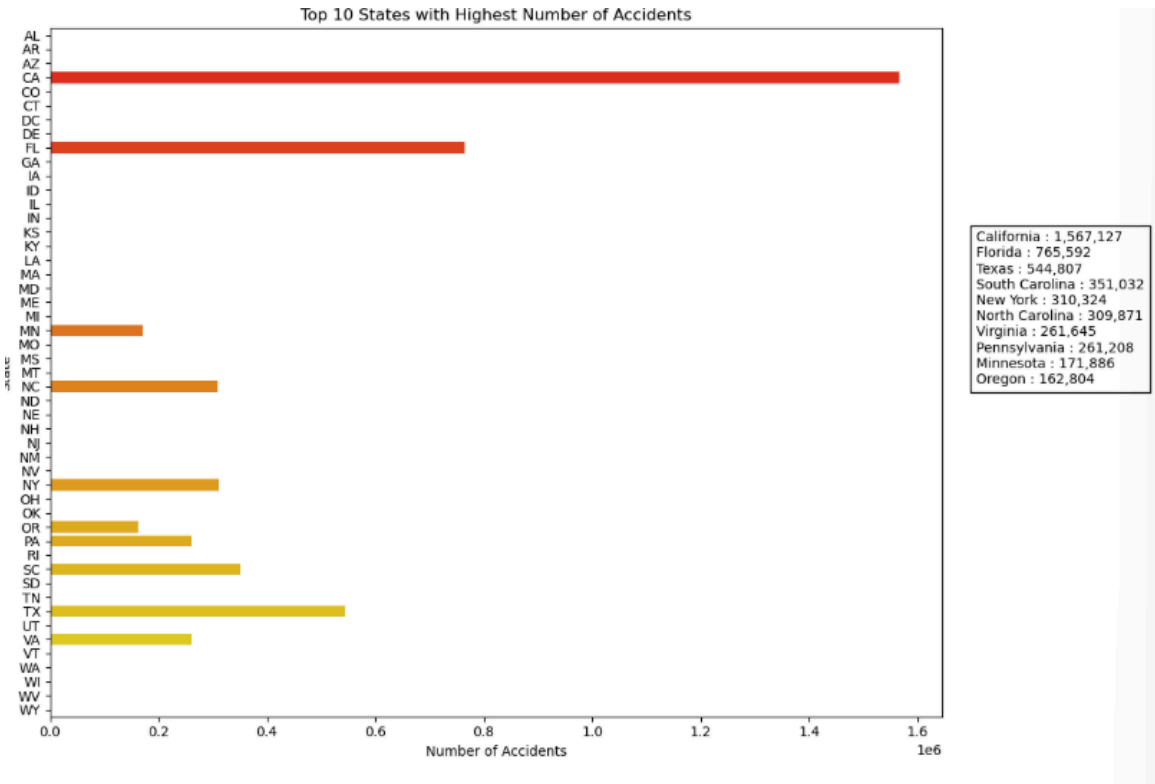

Figure 8 –Top 10 States with most accident

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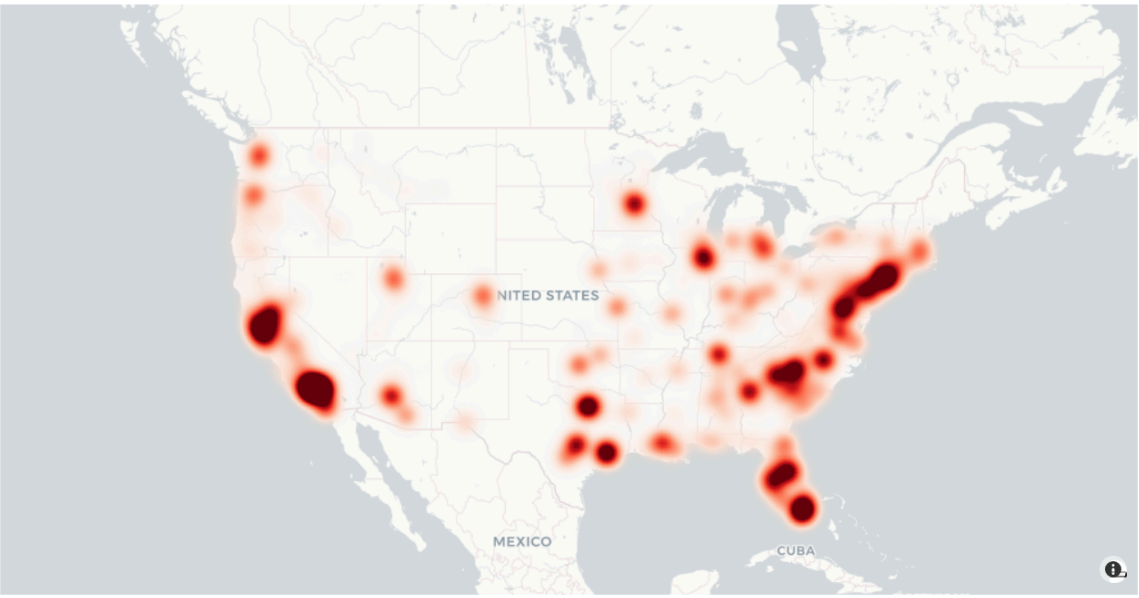

Figure 9 – Geographic Accident Hotspots (Heatmap)

Key insights:

1. Accident hotspots are heavily concentrated in major metropolitan regions, especially in states like South Carolina, Texas, and Florida, highlighting the impact of dense urban traffic and busy road networks.

2. Coastal & Major City Influence: Coastal regions and large urban hubs show higher accident density compared to central states, mainly due to higher population, heavy commuting, and extensive highway systems. Cities like Chicago and Detroit also form noticeable accident clusters.

3. Lower Density in Rural Regions: Central and sparsely populated states show very few accident hotspots, likely because of lower traffic volume and less congested road networks, indicating that accident frequency strongly relates to urbanization and traffic flow.

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Each visualization highlights patterns that help explain how traffic behavior, environmental conditions, and infrastructure contribute to accident risks.

Key findings from this project

• Most accidents occur during clear weather conditions, indicating that traffic volume plays a larger role than weather conditions.

• Accident frequency peaks during morning and evening commuting hours, reflecting higher traffic density during rush hours.

• Intersections, junctions, and traffic signals are high-risk locations due to increased vehicle interactions.

• Urban areas and highways show higher accident concentration due to heavy traffic flow.

Recommendations

1. Improve Safety at High-Risk Intersections
Deploy improved traffic signal timing, better road signage, and dedicated turning lanes to reduce accidents near junctions.

2. Implement Smart Traffic Monitoring
Use intelligent traffic management systems and congestion monitoring to reduce accident risk during peak traffic hours.

3. Target Infrastructure Improvements in Hotspot Areas
City planners should prioritize accident hotspot locations for improved road design, lighting, and safety infrastructure.

Project Outcome

This project demonstrates how large-scale traffic data can be analyzed using Python to uncover hidden patterns and accident risk factors.

The analysis identifies temporal accident trends, environmental influences, infrastructure risk areas, and geographic hotspots that contribute to accidents. These insights can support transportation authorities in making data-driven decisions to improve road safety.